

DUAL-CORE FIRMWARE, HOST SOFTWARE & RESULTS

STM32H747 MotionIPC Architecture, Live GUI, Telemetry, and Characterization

Customer-facing technical report
Revision 2.4 • 3 August 2026

Prepared by	Herder Elektronische Systemen
Primary subject	Dual-core firmware, host GUI, telemetry, logging, and measured system behavior
Evidence basis	MotionIPC source, motion firmware, GUI video, plots, logs, slide reviews, and technical reports
Distribution	Customer-facing; selected source excerpts and measured evidence included

DOCUMENT PURPOSE

This report presents the complete software side of the project: deep dual-core firmware, live operator software, data integrity, safety behavior, and measured results from the physical platform.

1. System Outcome

The completed system combines a custom STM32H747 dual-core IPC architecture with target-side motion control, encoder acquisition, host supervision, live visualization, structured logging, automated workflows, and postprocessing. The 400 MHz Cortex-M7 and 200 MHz Cortex-M4 communicate through OpenAMP/RPMsg for control traffic and a custom AXI-SRAM ring for motion records. The desktop application exposes the resulting system as an operable laboratory instrument rather than a firmware demonstration.

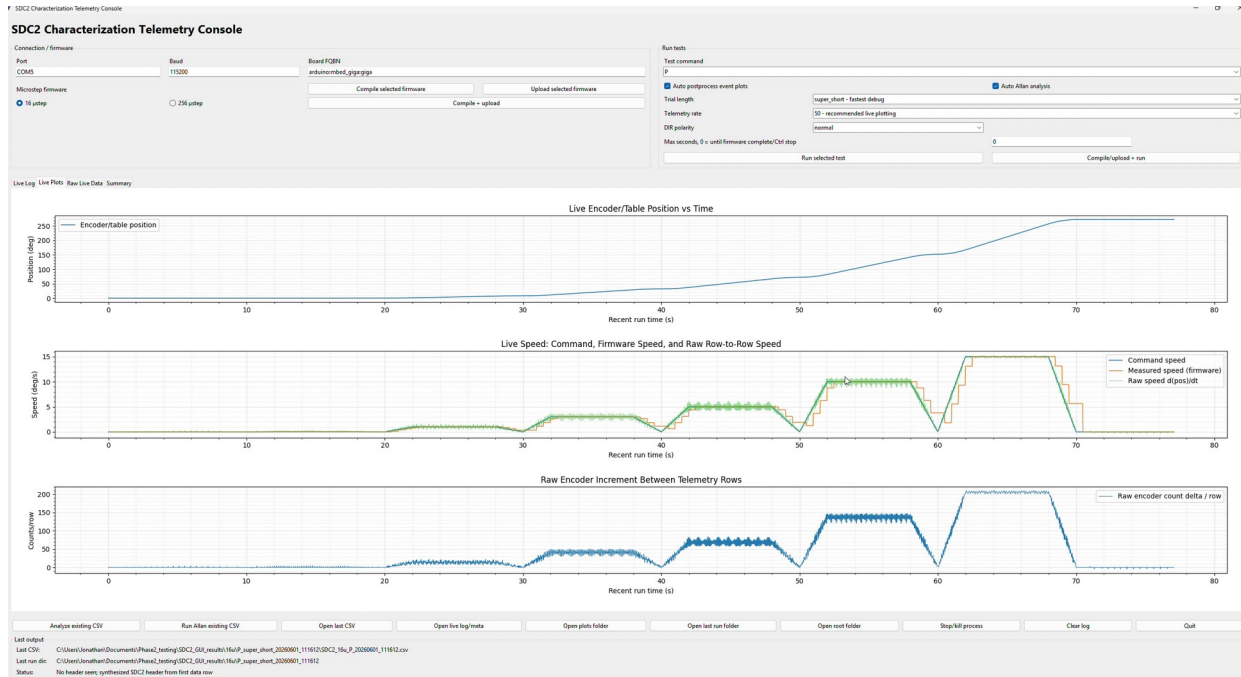


Figure 1. Operator-facing host application with connection controls, test selection, status, and live telemetry.

CUSTOMER-VISIBLE RESULT

An operator can connect the target, select a test, monitor command and encoder behavior, record timestamped data, recover interrupted runs, and launch analysis without controlling real-time motion from the PC.

2. Dual-Core Architecture

STM32H747 Dual-Core Motion Architecture

Control-plane messaging separated from high-rate data movement

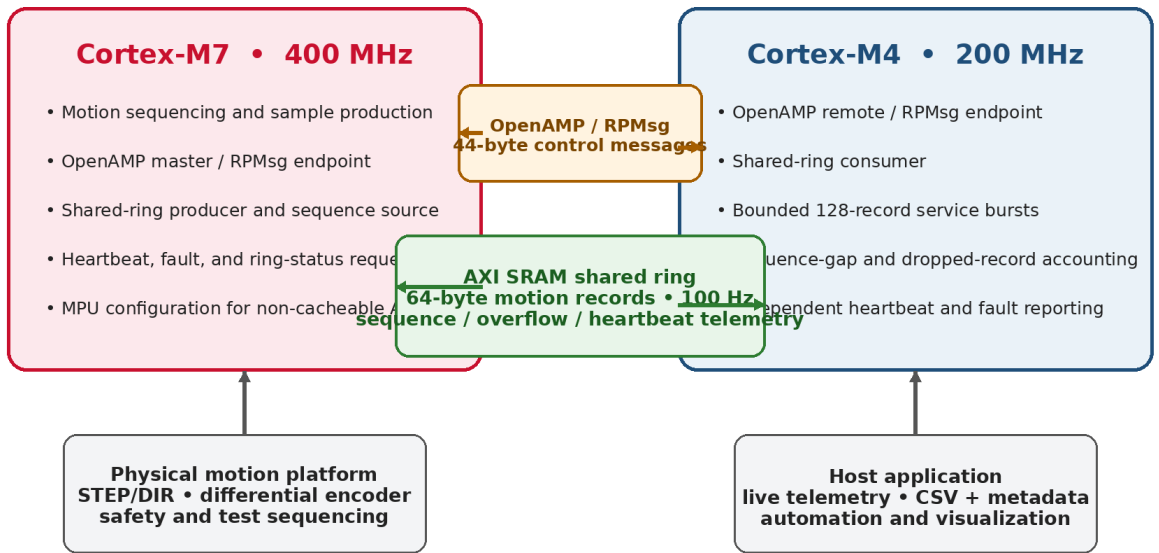


Figure 2. STM32H747 architecture separating the OpenAMP/RPMsg control plane from the AXI-SRAM motion-data path.

Verified Dual-Core Firmware Specifications

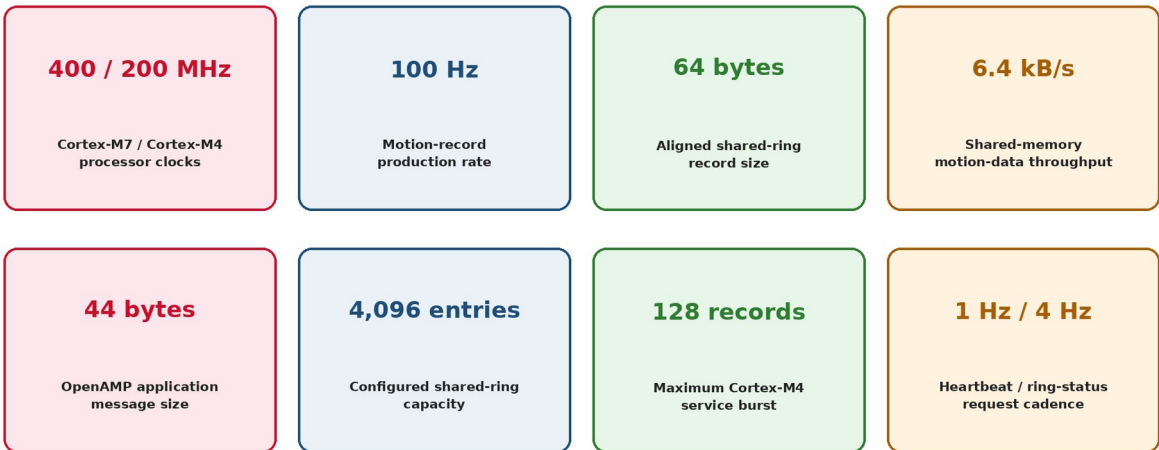


Figure 3. Source-verified processor, IPC, buffering, and service-cadence specifications.

2.1 Verified implementation table

Item	Verified implementation
Cores	400 MHz Cortex-M7 + 200 MHz Cortex-M4
Control IPC	OpenAMP/RPMsg; 44-byte application messages
Data IPC	256 KiB AXI-SRAM producer/consumer ring
Motion record	64 bytes, cache-line aligned
Record rate	100 Hz; 6.4 kB/s internal throughput
Consumer service	Up to 128 records per M4 service pass
Health cadence	1 Hz per-core heartbeat; 4 Hz ring-status requests

3. Core Ownership and Boot Sequence

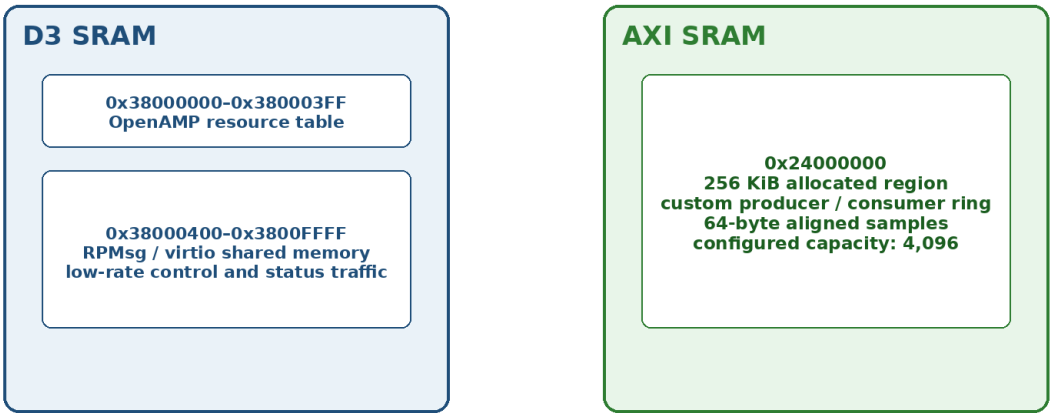
Cortex-M7 ownership	Cortex-M4 ownership
MPU/cache setup and HSEM release	Wake from STOP and initialize remote core
OpenAMP master and RPMsg endpoint	OpenAMP remote and RPMsg endpoint
Shared-ring producer and sequence source	Shared-ring consumer
Motion record production	Bounded 128-record processing bursts
Heartbeat and status requests	Gap, drop, heartbeat, and fault reports

3.1 Startup flow

Stage	M7	M4
1	Configure memory policy and clocks	Wait in STOP mode
2	Release HSEM0	Wake on HSEM event
3	Initialize producer	Initialize consumer
4	Create OpenAMP master endpoint	Create OpenAMP remote endpoint
5	Produce records and service messages	Consume records and service messages

MotionIPC Shared-Memory Map

Separate regions preserve ST OpenAMP resources while providing a high-throughput custom data path



Engineering review finding: 4,096 × 64-byte records plus metadata exceed the 256 KiB allocation by ~64 B

Figure 4. Shared-memory map and the identified ring-capacity correction.

4. IPC Protocol, Data Integrity, and Reliability

Mechanism	Implementation outcome
Sequence continuity	Monotonic sequence numbers and CM4 gap counting expose missing records.
Overflow visibility	Producer-side drop counter reports a full ring instead of silently overwriting data.
Independent liveness	Separate M7/M4 heartbeat counters distinguish core-local failures.
Fault propagation	RPMMsg fault messages and per-core flags carry diagnostic state.
Bounded workload	M4 consumes no more than 128 records per pass to preserve OpenAMP service.
Memory ordering	Barriers protect producer/consumer state transitions.
Cache coherency	M7 MPU marks the AXI ring non-cacheable.

REVIEW FINDING

The configured $4,096 \times 64$ -byte sample array fills the 256 KiB allocation before metadata is counted. A production release should use 4,095 entries or enlarge the region.

4.1 Motion record contents

Field group	Contents
Identity and time	Sequence number and microsecond timestamp
Motion state	Encoder count, position, commanded speed, measured speed, tracking error
Execution state	Target state, flags, and status fields

5. Target Motion Firmware

Real-time pulse generation, encoder capture, test progression, and deadman behavior remain on the embedded target. The host requests actions and records results but does not schedule motion.

Control or safety item	Verified value
STEP pulse high time	4 μ s
Step-rate safety cap	25,000 pulses/s
Maximum pulse backlog	25 pulses/pass
Loop-overflow warning	25 ms
Speed-estimation window	500 ms
Host heartbeat	500 ms
Target deadman timeout	2,000 ms
Position correction	20 pulse-Hz/degree; 250 Hz absolute cap; 5% relative cap

5.1 Encoder and drive scaling

Parameter	Value
Gear ratio	24.65:1
256-microstep pulses/table revolution	1,262,080
Command increment	$\sim 0.000285^\circ/\text{pulse}$
Encoder CPR	5,000
Effective table counts/revolution	246,500
Measured increment	$\sim 0.001460^\circ/\text{count}$

6. Host GUI and Live Operation

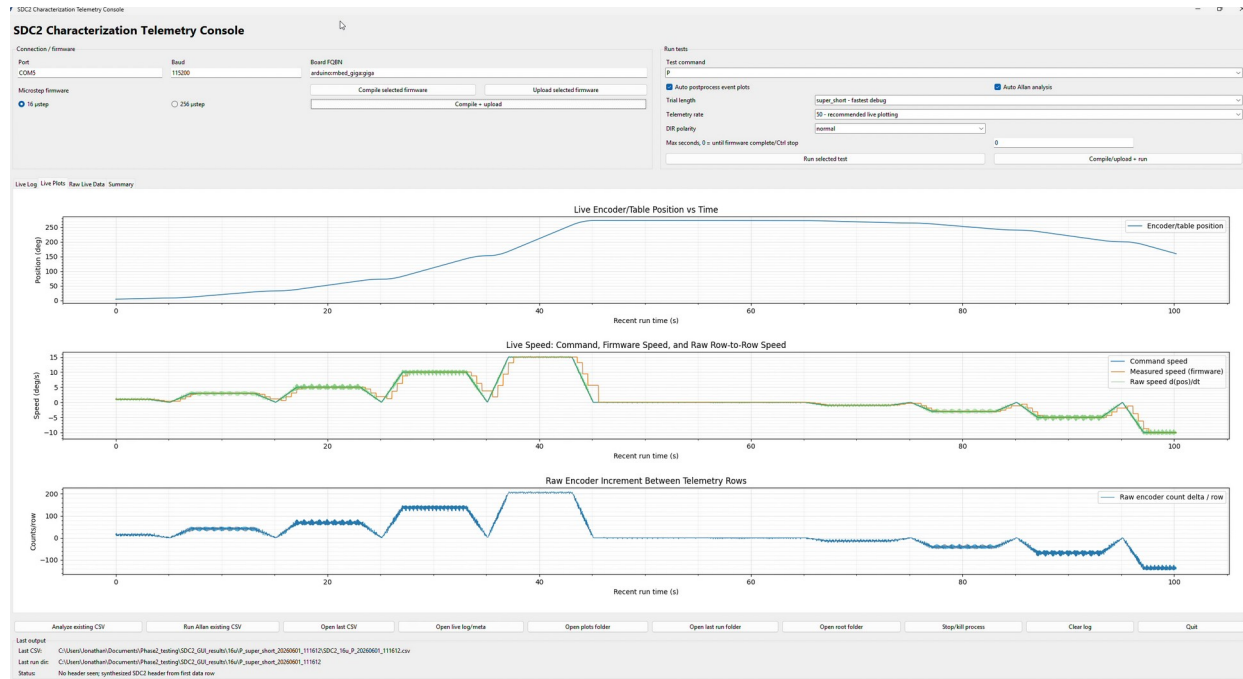


Figure 5. Active host-target test showing live command, measured motion, encoder position/rate, status, and logging state.

The GUI integrates connection management, command execution, test selection, live plots, file naming, logging state, and operator feedback. It provides one working surface for the embedded target rather than requiring separate serial terminals and analysis scripts.

- Live command-versus-measured speed and encoder-position plots.
- Raw encoder-rate visualization for diagnosing quantization and event timing.
- Target status, test progression, connection state, and completion markers.
- Automated filename generation, CSV capture, and separate metadata/comment retention.
- Launch and supervision of tracking, static, step, Allan, and sine-response workflows.

DEMONSTRATION ASSET

The repository includes the host-target interface video showing a scripted physical test progressing through positive and negative motion while telemetry and plots update live.

7. Telemetry, Logging, and Recovery

Capability	Implementation
Host transport	115,200-baud serial
Selectable output	10, 20, 50, or 100 Hz requests
Record schema	24 timestamped CSV fields
Validation	Header and row-width checks; malformed-row detection
Run completion	Test-specific completion markers
Metadata	Controller comments and run context stored separately
Recovery	Partial-run preservation after serial interruption
Safety	500 ms heartbeat and 2 s target deadman

7.1 Bandwidth result

A 115,200-baud 8-N-1 link provides an idealized 11,520 payload bytes/s. Representative 24-field ASCII rows can approach 189 bytes, limiting sustained full-record output to roughly 50–60 rows/s. The 100 Hz selection therefore identifies the need for shorter records, binary framing, or a faster external transport; the internal shared-memory path remains independent of this bottleneck.

ARCHITECTURE BENEFIT

High-rate inter-core motion data uses AXI SRAM rather than the external serial link. The host interface can evolve without redesigning the M7–M4 data plane.

8. Automated Test Workflow

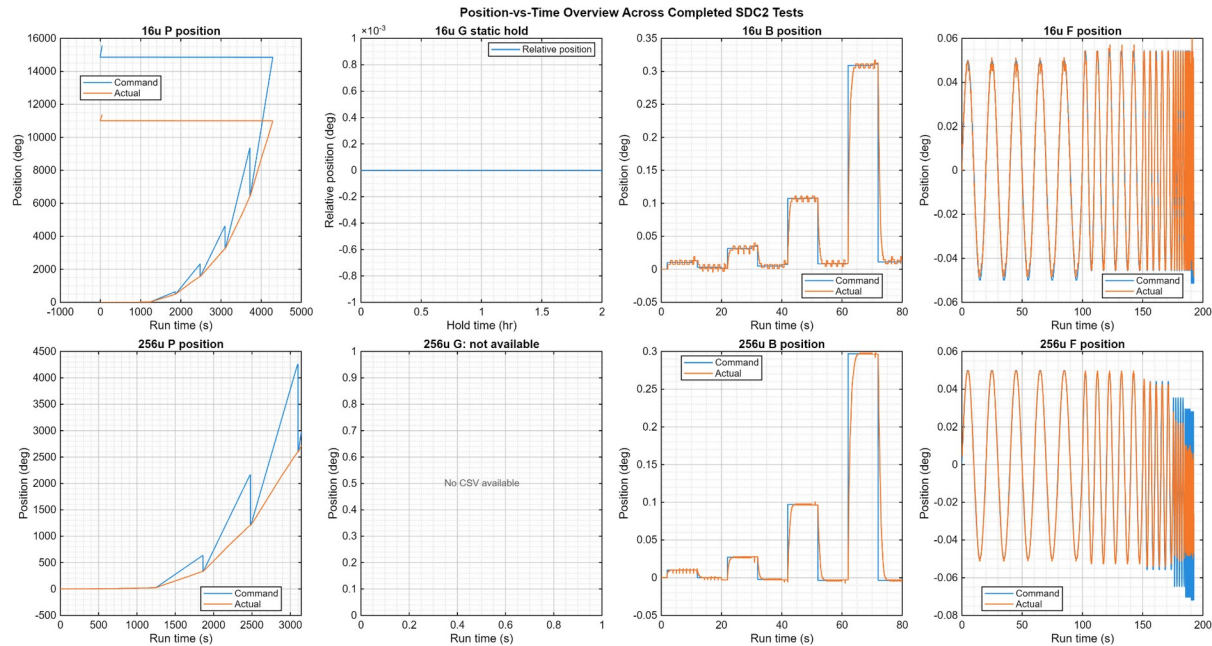


Figure 6. Representative scripted test sequences generated and recorded through the target/host workflow.

Workflow	Configured execution
Static hold	50 Hz × 180 s or 100 Hz × 90 s; 9,000 records
Dynamic tracking	100 Hz requested; 6,000 records per 60 s segment
Step response	Scripted command steps with rise, overshoot, settling, RMS, and peak-error analysis
Allan analysis	Static position and dynamic residual workflows
Sine response	Closed-loop command/actual response over approximately 0.33–2 Hz

Target-side sequencing keeps timing deterministic while the host application supervises, logs, validates, and postprocesses the resulting data.

9. Low-Speed Tracking Results

SDC2 ONLY | 256 MICROSTEP BASELINE

Speed tracking summary: usable ultra-slow range, clear high-speed ceiling

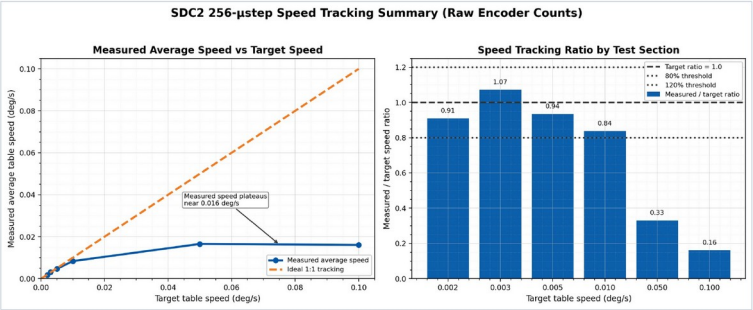


Figure 1. Clean summary view with legend moved off the data region. Average speed is computed from total raw encoder count delta over each section.

Main result

- 0.002-0.01 deg/s tracks in the correct range at the current 256 microstep setting.
- Best tracking occurs around 0.003-0.005 deg/s; 0.01 deg/s is usable but below target.
- 0.05 and 0.10 deg/s fail similarly and plateau near 0.016 deg/s actual speed.
- Interpretation: higher commanded speeds likely require lower microstepping and/or controller tuning.

Numerical baseline

- 0.002 → 0.001818 deg/s, ratio 0.909
- 0.003 → 0.003214 deg/s, ratio 1.071
- 0.005 → 0.004676 deg/s, ratio 0.935
- 0.010 → 0.008378 deg/s, ratio 0.838
- 0.050 → 0.016560 deg/s, ratio 0.331
- 0.100 → 0.016071 deg/s, ratio 0.161

SDC2 Encoder-Based Speed Tracking

Raw encoder-event analysis | no filtering or grouped averaging

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Figure 7. Measured low-speed performance in the 256-microstep configuration.

Command (deg/s)	Measured (deg/s)	Ratio / interpretation
0.002	0.001818	0.909
0.003	0.003214	1.071
0.005	0.004676	0.935
0.010	0.008378	0.838
0.050	~0.016	Configuration throughput ceiling
0.100	~0.016	Configuration throughput ceiling

RESULT

The 256-microstep setup is an ultra-slow configuration. Best tracking occurred near 0.003–0.005°/s; faster commands plateaued near 0.016°/s and require a different microstep or pulse-rate configuration.

10. Constant-Speed and Step-Response Results

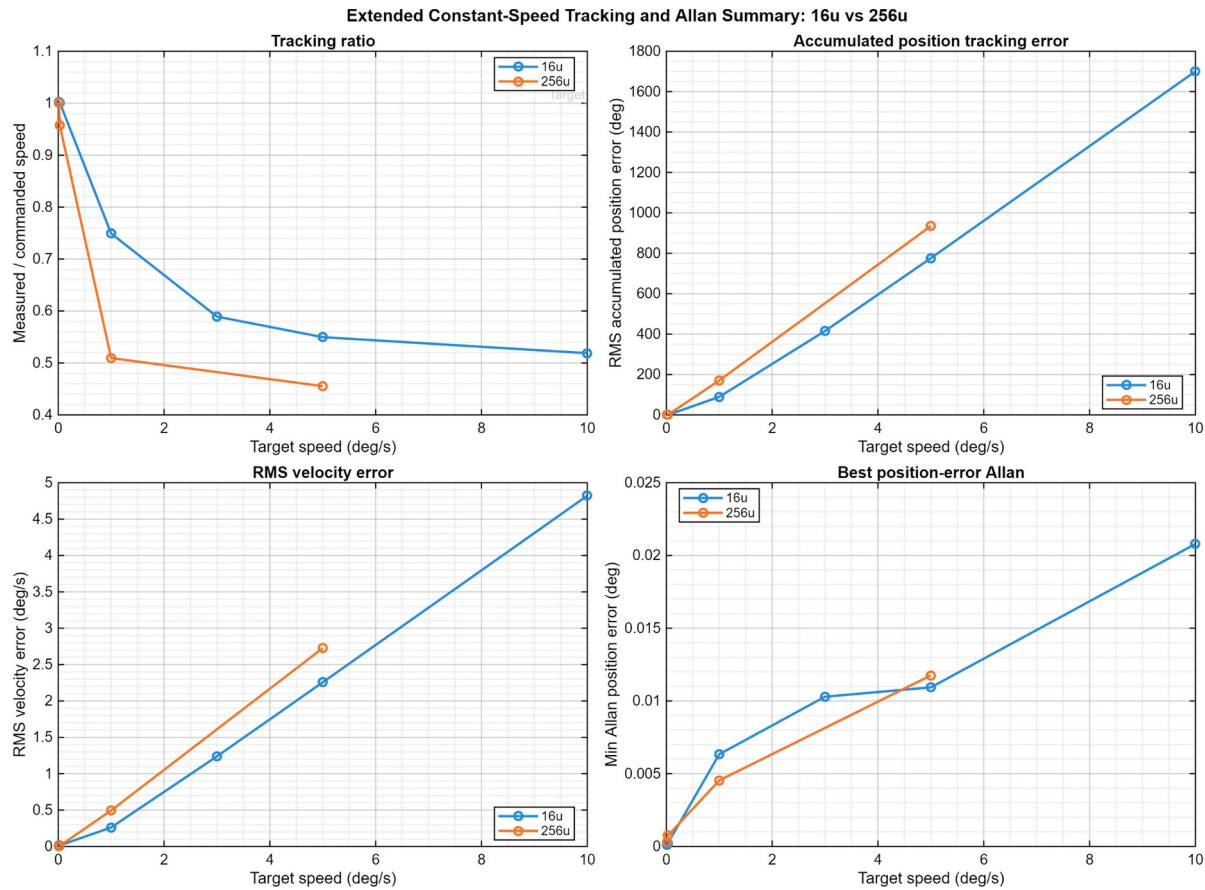


Figure 8. Representative constant-speed command and measured-motion tracking.

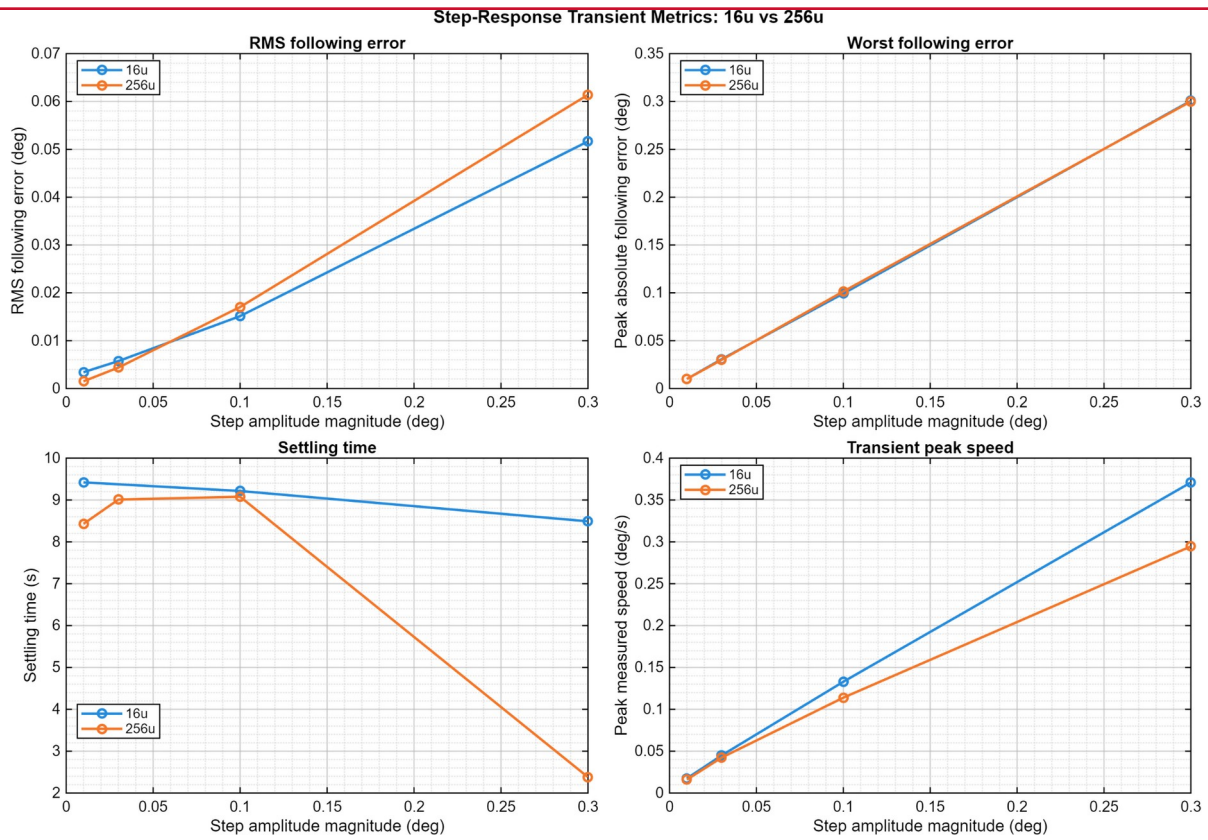


Figure 9. Representative scripted step-response characterization.

These results demonstrate that the embedded sequencing, timestamped acquisition, host logging, and analysis pipeline operated together on the physical platform. Step metrics are computed from retained records rather than from GUI rendering timestamps.

11. Closed-Loop Frequency-Response Results

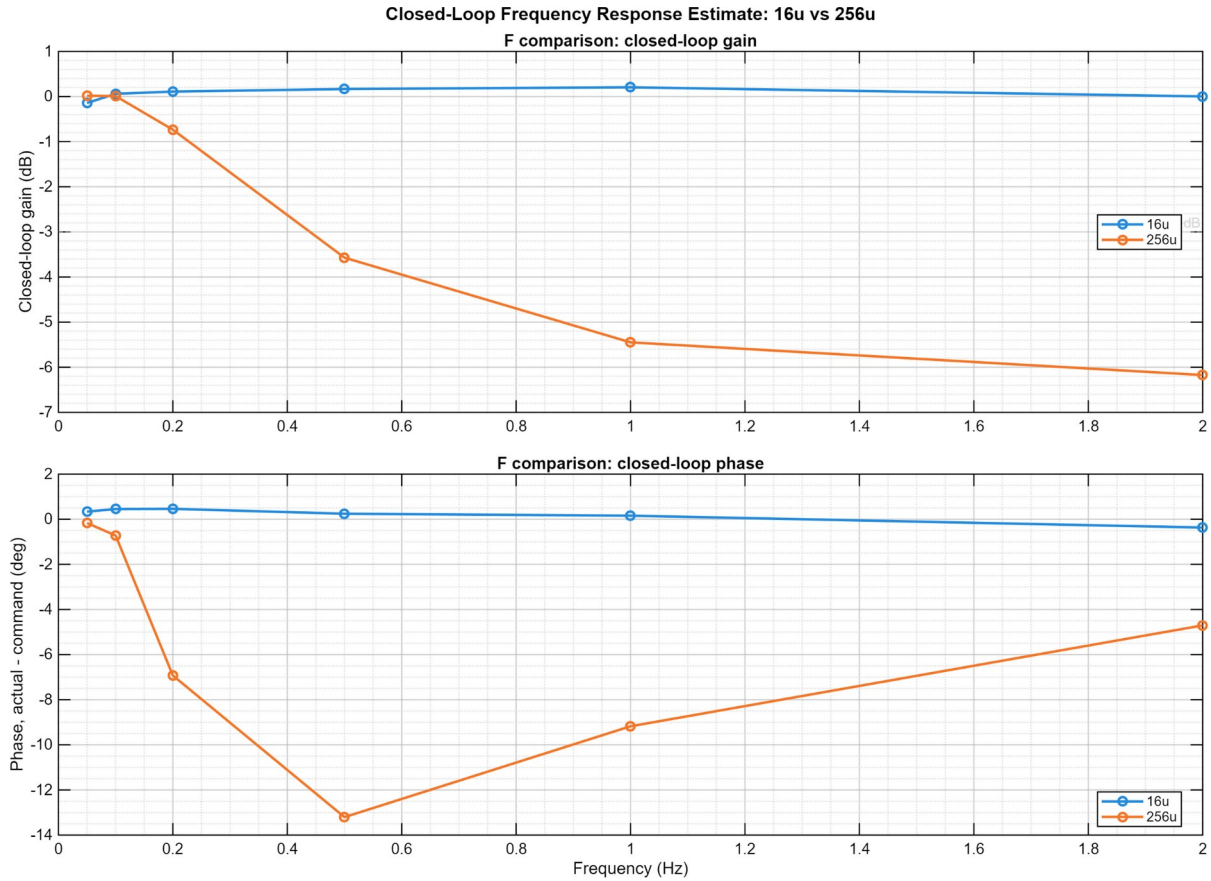


Figure 10. Closed-loop actual/command response over the tested sine-response range.

The workflow estimates gain and phase of measured motion relative to the commanded trajectory. It is a closed-loop response characterization, not a claim of open-loop plant identification or formal phase-margin measurement.

EVIDENCE SCOPE

The tested points span approximately 0.33–2 Hz. A precise interpolated –3 dB bandwidth is not claimed unless a validated crossing is present in the retained dataset.

12. Delivered Software Capabilities

Layer	Delivered capability
Dual-core firmware	OpenAMP/RPMsg coordination, shared-memory ring, HSEM startup, coherency policy, sequence/drop/fault accounting
Motion firmware	Pulse generation, encoder capture, correction, automated sequencing, heartbeat, deadman, and overrun monitoring
Host application	Connection, command execution, live plotting, test orchestration, logging, metadata, and recovery
Analysis	Low-speed, tracking, step response, Allan deviation, and closed-loop sine-response processing
Documentation	Architecture diagrams, FRDs, selected source excerpts, videos, plots, and public repository

13. Limitations and Release Actions

Item	Controlled statement
Ring allocation	Reduce capacity to 4,095 or reserve more than 256 KiB.
Integrated build	Tag and preserve one release binary/build proving final module integration.
IPC timing	Instrument end-to-end latency and worst-case jitter.
External telemetry	Use binary framing or faster transport for sustained 100 Hz full records.
Power	Measure idle, steady-state, and peak electrical power.
Reliability	Mechanisms are present; no MTBF percentage is claimed.

14. Attribution and Revision Control

Herder Elektronische Systemen developed the dual-core firmware architecture, motion-control integration, host software, logging and analysis workflows, interface electronics, and customer-facing documentation represented here. No third-party affiliation or endorsement is claimed.

Rev	Date	Description
2.4	2026-08-03	Complete firmware/software report integrating dual-core architecture, GUI, telemetry, reliability behavior, automated workflows, and measured results.

RELEASE GATE

Correct the ring allocation, preserve a tagged integrated build, measure IPC timing and electrical power, and validate sustained external transport before production release.